

Aarsh Desai

☎ +91 9727722530 ✉ desaiaarsh4@gmail.com in [LinkedIn](#) ○ [GitHub](#) 📁 [Portfolio](#)

Experience

Nanonets

July 2025 - Present

Deep Learning Engineer

Bengaluru, Karnataka

- Architected a multi-tiered agentic memory pipeline driven by real-time user feedback with the goal of eliminating instruction conflicts and enabling AI agents to dynamically adapt to evolving logic and edge-case rules.
- Fine-tuned and optimized domain-specific multimodal architectures, engineering robust training pipelines that minimized hallucinations and maximized extraction quality on highly noisy, real-world inputs.
- Led high-priority precision initiatives by systematically diagnosing failure cases, delivering tailored model improvements to consistently meet strict enterprise SLAs across diverse customer workflows.

Nanonets

January 2025 - July 2025

Deep Learning Intern

Bengaluru, Karnataka

- Investigated and fine-tuned cutting-edge foundation models (e.g., SigLIP, QWEN2.5-VL) for an image embedding usecase, leveraging contrastive learning techniques to optimize document representations for clustering and retrieval.
- Engineered a robust data pipeline for Human-in-the-Loop (HIL) annotation, integrating active-learning frameworks and custom annotation UIs to accelerate high-quality label acquisition by 70%.
- Designed a custom vision embedding model that outperformed SOTA architectures by 30% in document retrieval tasks across highly unstructured layouts and templates.

Robert Bosch Centre for Data Science and AI, IIT Madras

June 2024 - November 2024

Research Intern (under the mentorship of Dr. Gokul S Krishnan and Dr. Sanjay Bankapur)

Remote

- Engineered a multimodal stock prediction system, integrating transformer-based social media sentiment, financial news, and technical indicators to enhance swing trading strategies.
- Optimized time-series forecasting models and feature-selection pipelines, achieving a 25% improvement in prediction accuracy over baseline historical models.

Centre for Educational Technology, IIT Bombay

May 2023 - September 2024

Research Intern (under the mentorship of Dr. Ashwin TS and Dr. Ramkumar Rajendran)

Mumbai, Maharashtra

- Developed AffectBots: A web-based tutoring system using multimodal analysis (audio, video, text) to analyze 100+ hours of student engagement data with 88% accuracy in emotion recognition.
- Implemented speaker diarization using PyAnnote and Whisper Speech Recognition model for Indo-English collaborative learning environments, analyzing 140+ hours of student interactions and outperformed existing speaker diarization models by 65%

Vocab.Ai

August 2023 - August 2024

MLOps Intern

Remote

- Built a full-stack data pipeline processing 100,000+ customer service conversations via Google Business APIs with 98% ingestion reliability, deploying a web UI tracking key behavioral metrics.
- Fine-tuned Meta LLaMA 2 and OpenAI GPT-3.5 Turbo using LangChain for sentiment analysis, topic modeling, and conversation scoring. Improved model accuracy by 8% through hyperparameter tuning and data augmentation techniques.

Dept of CSE, NIT Puducherry

December 2023 - January 2024

Deep Learning Intern (under the mentorship of Dr. Girish GN and Dr. Sanjay Bankapur)

Karaikal, Puducherry

- Optimized deep learning models for lung nodule classification and segmentation on the LIDC-IDRI dataset, reducing energy consumption by 77% with near-zero degradation in accuracy.
- Implemented automated training/testing procedures and model optimizations (pruning, quantization) for lung nodule analysis, ensuring reproducible results while reducing memory and compute requirements.

Achievements

- **Director’s Gold Medal for Best Outgoing Student** – Awarded to the top all-around graduating student in the Batch of 2025 at IIIT Dharwad.
- **Best Student Paper Award** – Recognized at the IEEE TALE 2024 international conference for outstanding research contributions.
- **Global Rank 19** – Placed 19th globally out of thousands of competitive programmers in CodeChef Starters 57.
- **Winner** – Secured 1st place at the Ganglia Tech Collegiate Hackathon.
- **Top-Tier Competitive Programmer** – Achieved a peak 4-Star rating on CodeChef and ranked within the top 3% of global participants on LeetCode.

Research Publications

- **Desai, A.**, Kartik, N. V. J. K., Gupta, P., Vinayak, V., T S, A., Vanahalli, M. K., & Rajendran, R. (2024, December). Advancing Speaker Diarization With Whisper Speech Recognition for Different Learning Environments. *2024 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE 2024)*. Bengaluru, India. DOI:10.1109/TALE62452.2024.10834319
- Kartik, N. V. J. K., Gupta, P., M, V., **Desai, A.**, Badhe, V., T S, A., Vanahalli, M.K., & Rajendran, R. (2024, December). Unlocking the Triggers: Automating the identification of Triggers of Socially Shared Metacognitive Regulation in Collaborative Problem-Solving. *The International Conference on Technology 4 Education (T4E 2024)*. Gandhinagar, India. DOI:10.1007/978-981-95-1734-3_6
- Sharma, V., Kartik, N. V. J. K., Vinayak, **Desai, A.**, Gupta, P., Vanahalli, M. K., Prasad, B. M., Bankapur, S. S., & Girish, G. N. (2025, August). Energy-Efficient Depthwise Separable CNNs for Classification and Segmentation of Lung Cancer Nodules. *6th IEEE India Council International Subsections Conference (INDISCON 2025)*. Rourkela, India. DOI:10.1109/INDISCON66021.2025.11254247
- **Desai, A.**, Kartik, N. V. J. K., Gupta, P.,M., Vinayak, Vanahalli, M. K., Rajendran, R., T S, A. & Biswas, G.(2025, December). Exploring Emotional Trajectories through Valence and Arousal in Open-Ended Math Learning Environments. *The International Conference on Technology 4 Education (T4E 2025)*. Chennai, India.
NSF Award:2112635

Technical Skills

AI & Deep Learning: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, OpenAI API, OpenCV, Keras
Model Optimization & Serving: Quantization, Pruning, vLLM, FastAPI, Flask, Triton Inference Server
Core Expertise: Agentic AI, Vision-Language Models (VLMs), Multimodal AI, Document AI, MLOps
Languages & Databases: Python, C/C++, SQL, Golang, PostgreSQL, MongoDB, Redis, Cassandra, Elasticsearch
Cloud, DevOps & MLOps Tools: AWS, GCP, Docker, Kubernetes, Git, GitHub Actions, Argo CD, Grafana

Education

Indian Institute of Information Technology, Dharwad <i>BTech in Data Science and Artificial Intelligence (Cumulative GPA: 9.19)</i>	May 2025 Dharwad, Karnataka
G.D. Goenka International School <i>CBSE HSC (Percentage: 94.4)</i>	May 2021 Surat, Gujarat

Projects

Online Schooling for the Specially Abled <i>TensorFlow, Keras, OpenCV, Streamlit, Docker, AWS</i> 	May 2023
• Engineered a real-time computer vision pipeline utilizing OpenCV and TensorFlow to analyze student facial expressions and gaze metrics, enabling dynamic content adaptation for accessible learning. • Containerized the application using Docker and deployed it on AWS (EC2/S3) infrastructure to ensure scalable, low-latency processing of video stream data. • Developed the end-to-end cloud-native platform within a 48-hour sprint, securing 1st Place out of 50 teams at the collegiate hackathon for innovation in accessible education.	